

AP Chemistry Assignment: Atomic Structure, Emission Spectra, and Precipitation Reactions

Name: _____

Date: _____

Class Period: _____

Directions

- Show all work for calculations.
- Include units and appropriate significant figures.
- For net ionic equations, include physical states.
- Use the following equations as needed:

$$E_n = -\frac{2.18 \times 10^{-18} \text{ J}}{n^2}$$

$$\Delta E = E_{\text{final}} - E_{\text{initial}}$$

$$c = \lambda \nu$$

$$E = h\nu = \frac{hc}{\lambda}$$

where:

$$c = 3.00 \times 10^8 \text{ m/s}$$

$$h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$$

Part I: Bohr Model and Electron Transitions

1. Comparative Energy Levels

In the Bohr model of the hydrogen atom, the energy required to excite an electron from $n = 2$ to $n = 3$ is:

- A. Less than the energy required to excite an electron from $n = 3$ to $n = 4$
- B. Equal to the energy required to excite an electron from $n = 3$ to $n = 4$
- C. Greater than the energy required to excite an electron from $n = 3$ to $n = 4$
- D. Unrelated to the energy required for the $n = 3$ to $n = 4$ transition

Answer: _____

Explain your reasoning.

2. Energy of an Excitation

Calculate the energy required to excite a hydrogen electron from $n = 2$ to $n = 3$.

Work:

Answer: _____ J

3. Comparing Electron Transitions

Consider the following transitions in a hydrogen atom:

- I. $n = 2 \rightarrow n = 1$
- II. $n = 3 \rightarrow n = 1$
- III. $n = 1 \rightarrow n = 4$

Which transition releases the greatest amount of energy?

- A. I only
- B. II only
- C. III only
- D. I and II

Answer: _____

Explain your reasoning.

4. Longest Wavelength

When an electron drops to $n = 1$, which transition produces the photon with the longest wavelength?

$$n = \boxed{} \rightarrow n = 1$$

Explain your reasoning.

5. Energy, Frequency, and Wavelength

An electron in a hydrogen atom drops from $n = 3$ to $n = 2$.

a. Is energy absorbed or released?

Answer: _____

b. Is the emitted photon's wavelength relatively long or short compared with a photon emitted during a $n = 2 \rightarrow n = 1$ transition?

Answer: _____

c. Explain your answer.

6. Numerical Transition Analysis

Calculate the wavelength of light emitted when a hydrogen electron drops from $n = 3$ to $n = 2$.

Work:

Answer: _____ nm

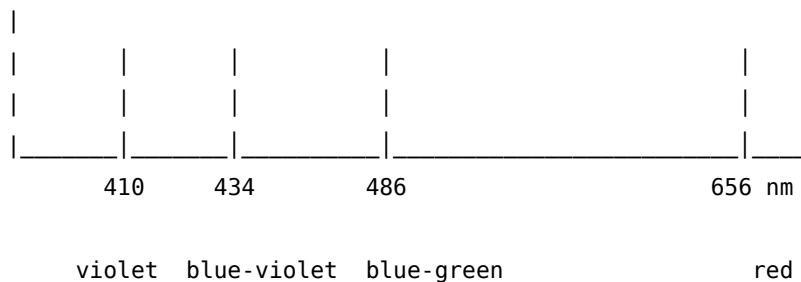
Approximate color: _____

Part II: Emission Spectra

Hydrogen Emission Spectrum

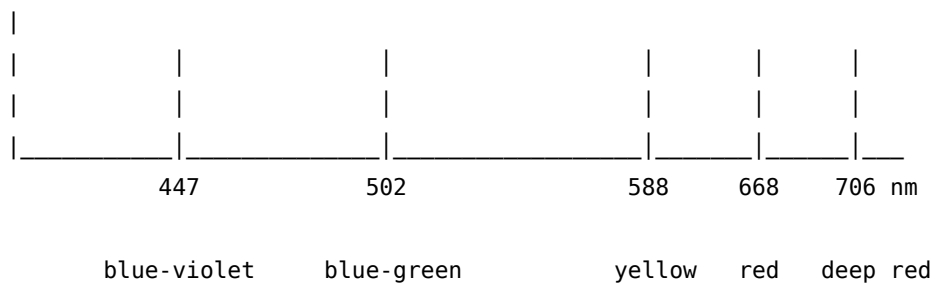
Use the simplified hydrogen spectrum below.

Relative intensity



Helium Emission Spectrum

Relative intensity



7. Spectrum Identification

Which spectrum is represented by the first diagram?

Answer: _____

Which spectrum is represented by the second diagram?

Answer: _____

8. Greatest Photon Energy

Which hydrogen emission line has the greatest energy?

- A. 410 nm
- B. 434 nm
- C. 486 nm
- D. 656 nm

Answer: _____

Explain your reasoning using wavelength and frequency.

9. Lowest Photon Energy

Which helium emission line has the lowest energy?

Answer: _____

Explain your reasoning.

10. Hydrogen Spectrum Calculation

Calculate the energy of the hydrogen emission line at 656 nm.

Work:

Answer: _____ J

11. Helium Spectrum Calculation

Calculate the frequency of the helium emission line at 588 nm.

Work:

Answer: _____ Hz

12. Identifying a Mystery Gas

A mystery gas produces bright emission lines at:

447 nm, 502 nm, 588 nm, 668 nm, 706 nm

Is the gas more likely hydrogen or helium?

Answer: _____

Justify your answer using evidence from the spectra.

13. Mixture Spectrum

A sample contains both hydrogen and helium. Predict how the emission spectrum of the mixture would compare with the spectra of pure hydrogen and pure helium.

Part III: Dalton's Atomic Theory

14. Historical Atomic Theory

Which statement was **not** one of Dalton's original hypotheses?

- A. Matter is composed of atoms.
- B. Atoms of the same element are identical in mass and properties.
- C. Atoms can be divided into protons, neutrons, and electrons.
- D. Compounds form when atoms combine in whole-number ratios.

Answer: _____

15. Atomic Models

Explain one major difference between Dalton's model of the atom and the modern atomic model.

Part IV: Net Ionic Equations

Write the balanced net ionic equation for each reaction. Include physical states.

16. Copper(II) Iodide and Lithium Carbonate

Aqueous copper(II) iodide and lithium carbonate are mixed.

Net ionic equation:

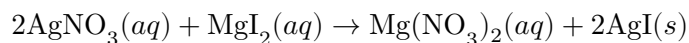
17. Lead(II) Nitrate and Potassium Iodide

Aqueous lead(II) nitrate and potassium iodide are mixed.

Net ionic equation:

18. Silver Nitrate and Magnesium Iodide

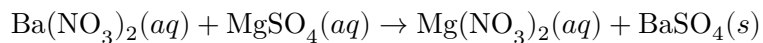
The following reaction occurs:



Net ionic equation:

19. Barium Nitrate and Magnesium Sulfate

The following reaction occurs:



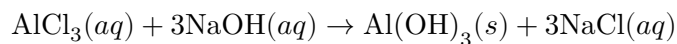
Net ionic equation:

Part V: Precipitation Stoichiometry

20. Aluminum Chloride Analysis

A 60.00 mL sample of aluminum chloride solution is treated with excess sodium hydroxide. The aluminum hydroxide precipitate has a mass of 0.780 g.

The reaction is:



Calculate the molarity of the original AlCl_3 solution.

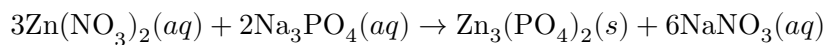
Work:

Answer: _____ M

21. Zinc Nitrate Analysis

A 45.00 mL sample of zinc nitrate solution reacts with excess sodium phosphate. The zinc phosphate precipitate has a mass of 0.579 g.

The reaction is:



Calculate the molarity of the original $\text{Zn}(\text{NO}_3)_2$ solution.

Work:

Answer: _____ M

Part VI: AP-Style Free Response

22. Connecting Spectra and Energy

A student observes that hydrogen produces several bright lines rather than a continuous spectrum.

- a. Explain why the hydrogen spectrum contains discrete lines.

- b. Explain how the spectrum supports the idea that electron energies are quantized.

- c. What happens to photon energy when the emitted wavelength decreases?

23. Experimental Analysis

Two unknown gases produce the following emission lines:

Gas	Emission Lines
X	410 nm, 434 nm, 486 nm, 656 nm
Y	447 nm, 502 nm, 588 nm, 668 nm, 706 nm

- a. Identify Gas X and Gas Y.

Gas X: _____

Gas Y: _____

- b. Which gas produces the highest-energy observed photon?

Answer: _____

- c. Justify your answer.

- d. Explain how emission spectroscopy could be used to identify elements in a star.

Teacher Answer Key

1. C. Greater than. The energy levels become closer together as n increases, so the $n = 2 \rightarrow 3$ transition requires more energy than $n = 3 \rightarrow 4$.
2. 2.18×10^{-19} J
3. B. II only. The $n = 3 \rightarrow 1$ transition has the largest energy decrease. Transition III is an absorption process.
4. $n = 2$. The $n = 2 \rightarrow 1$ transition has the smallest energy change among transitions ending at $n = 1$, so it emits the longest wavelength.
5. Energy is released. The $n = 3 \rightarrow 2$ transition has a smaller energy difference than the $n = 2 \rightarrow 1$ transition, so it produces a longer-wavelength photon.
6. Approximately 656 nm; red light.
7. First spectrum: hydrogen. Second spectrum: helium.
8. A. 410 nm. The shortest wavelength has the greatest frequency and energy.
9. 706 nm. It is the longest wavelength listed.
10. 3.03×10^{-19} J
11. 5.10×10^{14} Hz
12. Helium. The listed wavelengths match the helium spectrum.
13. The mixture would show the combined emission lines of both hydrogen and helium.
14. C. Atoms can be divided into protons, neutrons, and electrons.
15. Dalton proposed that atoms were indivisible. The modern model shows that atoms contain subatomic particles and have an internal structure.
16. $\text{Cu}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{CuCO}_3(\text{s})$
17. $\text{Pb}^{2+}(\text{aq}) + 2\text{I}^{-}(\text{aq}) \rightarrow \text{PbI}_2(\text{s})$
18. $\text{Ag}^{+}(\text{aq}) + \text{I}^{-}(\text{aq}) \rightarrow \text{AgI}(\text{s})$
19. $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
20. 0.167 M
21. 0.100 M
22. Only specific wavelengths are emitted because electrons can occupy only specific energy levels. Each line corresponds to a specific energy difference between allowed levels. Decreasing wavelength means increasing frequency and increasing photon energy.

23. Gas X: hydrogen. Gas Y: helium. Gas X produces the highest-energy observed photon because 410 nm is shorter than Gas Y's shortest listed wavelength, 447 nm. Emission spectra can be compared with known spectra to identify elements in stars.